
Causal Model Provenance via Shard Membership Auditing

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Abstract

1 Foundational models are increasingly adapted for downstream applications, pro-
2 ducing large number of derivative models. This motivates the *model provenance*
3 problem: Given a target model and a candidate parent model, can one test, with
4 statistical guarantees, whether the target was derived from the parent? Reliable an-
5 swers are essential for intellectual property disputes, vulnerability tracing, and ac-
6 countability, yet existing tests rest on evidence that is not causally motivated. We
7 address this gap by formulating model provenance as a causal inference problem,
8 making the lineage relation explicit and carefully analyze the underlying assump-
9 tions. We propose a general statistical framework for black-box model provenance
10 that reduces the problem to data membership auditing. We test whether the tar-
11 get model inherits knowledge of a secret shard such as a subset of training data
12 or licensed corpus from the parent candidate model. This reduction yields prove-
13 nance tests with rigorous statistical guarantees. We instantiate our framework for
14 LLM/ViT provenance, testing whether a target LLM/ViT was fine-tuned from a
15 candidate parent, and demonstrate its effectiveness across a diverse set of models
16 and datasets.

17 1 Introduction

18 Foundation models have become reusable building blocks for modern machine-learning systems.
19 Rather than training models from scratch, practitioners increasingly start from public checkpoints,
20 proprietary base models or open-source model families, and then adapt them through methods
21 such as fine-tuning, distillation or model merging. This pervasive reuse has motivated the *model*
22 *provenance* problem: Given a target model and a candidate parent, can we determine whether
23 the target was derived from the parent? This lineage relation matters for intellectual-property dis-
24 putes, license compliance, vulnerability tracing, data-governance obligations, scientific accountabil-
25 ity [5, 12, 18, 28]. In practice, auditors typically have only black-box query access to both models.

26 Recent works approach black-box provenance statistically [11, 17]. For instance, Nikolic et al. [17]
27 compare output similarity between target and parent against a set of independently trained control
28 models, while Kuditipudi et al. [11] test whether the target depends on randomized traces of the
29 parent’s training run. However, the evidence these methods produce is not always aligned with what
30 makes provenance matter. Output similarity is sensitive to control set choice and to incidental factors
31 like shared inductive bias or overlapping pretraining corpora. Training order is informative when
32 transcripts are available, but the order itself is not the inheritance one wants to establish. In most
33 provenance disputes, the central question is whether the target inherited specific information carried
34 by the parent such as a private shard, curated corpus, licensed subset, or domain-specific training
35 knowledge. This raises a natural question: *Can black-box model provenance be statistically tested*
36 *through the information whose transmission makes provenance consequential?*

37 **Our approach.** We answer affirmatively by recasting provenance as a causal inference problem.
 38 The parent-target lineage relation is confounded by background factors such as architecture, opti-
 39 mization recipes, shared pretraining corpora. These are typically unobserved, rendering the deriva-
 40 tion relationship unidentifiable [27] from black-box observations alone. To address this challenge,
 41 we propose using a *secret shard* S whose transmission would be provenance-relevant, e.g., a private
 42 subset or licensed corpus. Under standard structural conditions, the shard acts as an instrumental
 43 variable [1], i.e., influence of the shard on the target must flow through the parent. We thus reduce
 44 black-box provenance to a *data membership auditing* problem on the shard and the target.

45 This reduction yields a hypothesis test with finite-sample guarantees. We cast the problem as distin-
 46 guishing a real shard split from an independent control split drawn from the same population, and
 47 quantify the power of any black-box distinguisher by its *shard advantage*. Under the causal model
 48 in which the target could not have accessed the shard except through the parent, rejection of the null
 49 hypothesis implies, with high probability, that the parent-target lineage exists, with the false-positive
 50 rate bounded by design. We instantiate the framework for LLM provenance, testing whether a target
 51 LLM was fine-tuned from a candidate parent. Across diverse base models, fine-tuned derivatives,
 52 and shard datasets, the test reliably detects derivation while controlling Type-I error in matched
 53 nonmember-vs.-nonmember controls.

54 **Contributions.** Our contributions are as follows.

- 55 • **Causal model provenance.** We formalize the model provenance problem in a causal model
 56 (Model 1, Definition 1), and identify the auditor-chosen secret shard as an instrumental variable
 57 that makes the relevant causal effect identifiable from black-box queries.
- 58 • **Shard-membership auditing with guarantees.** We reduce provenance testing to a shard distin-
 59 guishing game (Definition 4), and define the shard advantage. We give an empirical advantage es-
 60 timator that concentrates at the $O(1/\sqrt{m})$ rate and a thresholding test (Lemma 1) with a bounded
 61 Type-I error, leading to a provenance guarantee under the causal model (Theorem 1), where m is
 62 the held-out sample size in each shard.
- 63 • **An instantiation for LLM provenance.** We instantiate the framework for fine-tuned LLM and
 64 Vision Transformer targets, plug standard membership-inference scores into our distinguisher, and
 65 demonstrate the effectiveness of the test across a diverse set of models and datasets.

66 2 Related Work

67 We review most closely related works here, but provide more information in Appendix A.

68 **Model provenance.** The most directly related works are statistical tests for whether a target model
 69 was derived from a candidate parent. Nikolic et al. [17] test for derivation by comparing the out-
 70 put similarity between a target model and a candidate parent against a set of control models. The
 71 validity of the resulting evidence depends on whether the chosen control set faithfully represents
 72 the no-lineage null, and on factors such as shared inductive bias or overlapping pretraining corpora.
 73 Kuditipudi et al. [11] formulate provenance as palimpsestic membership inference. However, the
 74 procedure does not yield a fixed false-positive rate in the natural query setting where the auditor
 75 only has black-box access. We view black-box model provenance through a causal model, where
 76 the parent and the target are both causally influenced by a secret shard, which is the source of the
 77 provenance signal. Unlike prior work, we show that a distinguisher can be used to test for prove-
 78 nance with finite-sample guarantees on the false positive rate, without relying on control models
 79 or on the randomization of the parent’s training process. In our framework, the randomly shuffled
 80 training order [11] is a particular instantiation of the exogenous secret shard determining the parent.

81 **Membership inference attacks (MIA).** A growing line of work studies whether a given example
 82 was part of an LLM’s training data [3, 9, 14, 25, 34, 36]. Several works have, however, called the
 83 practical effectiveness of these attacks into question [7, 15]. The argument is that MIA evaluations
 84 frequently confound true membership with distributional shift and therefore cannot, on their own,
 85 prove that a model was trained on a specific datum. While these MI scoring methods can be plugged
 86 into our distinguisher, our contribution is a statistically grounded estimator. Our estimator bypasses
 87 the distributional-shift confounds highlighted by MIA literature [7, 15].

88 **Model fingerprinting & watermarking.** Model fingerprinting aims to identify a specific model,
 89 while watermarking schemes embed an ownership signal. Active schemes embed an identifying
 90 signal such as instruction-following backdoors in LLMs [21, 30, 32], or watermarks in the training
 91 data [23, 24]. Passive schemes instead read out signatures from the trained model, e.g., adversarial
 92 examples [13, 22, 31], parameter-direction or representation-level features [35], but typically require
 93 white-box access [33] and were developed for classifiers, transferring poorly to generative models.
 94 Service-level fingerprinting [19] is adjacent but answers a different question, i.e., which LLM a
 95 black-box API serves. Our test needs neither training-time modifications nor model weights. We
 96 select a secret shard and query the candidate, which makes it applicable to already-deployed models.

97 3 Our Framework

98 We consider the model provenance problem of determining whether a target model T is derived
 99 from a candidate parent model P . If T is derived from P , then we say that P causes T , denoted by
 100 $P \rightarrow T$. Otherwise, any observed dependence between P and T may be due to background factors,
 101 such as architectural choices, optimization recipes, or public pretraining corpora. In this case, P is
 102 not a causal parent of T , denoted by the absence of the edge $P \rightarrow T$.

103 We formalize model provenance as a causal inference problem, stated in Definition 1. The goal
 104 is to determine whether the candidate parent P is a direct causal parent of the target model T . In
 105 principle, the natural causal object is the interventional distribution

$$P(T \mid \text{do}(P)),$$

106 which asks whether intervening on the candidate parent changes the distribution of the target model.
 107 However, this object is generally not identifiable. This is because the background factors C , such
 108 as architecture, training recipe, or overlapping public corpora, are confounding factors that are not
 109 observed from black-box observations of (P, T) . In other words, the backdoor path [20] $P \leftarrow$
 110 $C \rightarrow T$ cannot be blocked by any observable adjustment set when C is latent, so $P(T \mid \text{do}(P))$ is
 111 in general *unidentifiable* from observational data on (P, T) [27].

112 **Definition 1** (Causal model provenance). Given a structural causal model (SCM), we say that T is
 113 derived from P if P is a direct causal parent of T , i.e., $P \rightarrow T$.

114 To obtain an identifiable causal witness for provenance, we introduce a shard-mediated causal
 115 model, stated in Model 1. The key idea is to use a secret shard S that affects the candidate parent
 116 P , but can affect the target T only through the candidate provenance edge $P \rightarrow T$. Thus, instead
 117 of attempting to identify the full causal effect of P on T , we test whether shard-specific information
 118 can be detected in T . Under the structural conditions below, any such dependence between S and T
 119 is attributable to transmission along the path $S \rightarrow P \rightarrow T$.

120 **Model 1** (Shard-mediated causal model). *Let S be the secret shard, P be the candidate parent*
 121 *model, T be the target model, and C denotes background factors not specific to the shard. The audit*
 122 *is interpreted with respect to a structural causal model (SCM) over (S, P, T, C) , which contains the*
 123 *edge $S \rightarrow P$, and the edge $P \rightarrow T$ may be active. The background factor C may also have edges*
 124 *$C \rightarrow P$ and $C \rightarrow T$, but these are unobserved. The model satisfies the following conditions:*

- 125 1. S is a direct causal parent of P .
- 126 2. S is not a direct causal parent of T .
- 127 3. **No shard-specific confounding.** *There is no latent factor H that creates dependence between*
 128 *S and T while bypassing P . Formally, S is exogenous¹ and there are no paths of the form*
 129 *$S \leftarrow H \rightarrow T$.*

130 In our audit setting, the secret shard S can be a subset of the training data of the candidate parent P ,
 131 known only to the auditor. We assume the shard S is private, even though the underlying population
 132 $\mathcal{D}$ from which it is drawn may be public or partially known. Formally, let $\mathcal{X}$ be some domain and
 133 let $\mathcal{D}$ be a distribution on $\mathcal{X}$. A *secret shard* is a random variable

$$S = (x_1, \dots, x_n) \sim \mathcal{D}^n.$$

¹Exogeneity of S is the claim that S has no causes inside the model.

Under Model 1, the secret shard S plays the role of an instrumental variable for detecting whether the edge $P \rightarrow T$ transmits shard-specific information [1, 20]. The three conditions in Model 1 correspond to the standard IV requirements: (i) *relevance*, S is a direct causal parent of P , so the parent model carries shard-specific information; (ii) *exclusion*, the only directed path from S to T is $S \rightarrow P \rightarrow T$, so any influence of S on T must transit through P ; and (iii) *independence of the instrument*, the no-shard-specific-confounding clause rules out the path $S \leftarrow H \rightarrow T$. Combined with the exogeneity $S \sim \mathcal{D}^n$, these conditions ensure that S is unconfounded with C .

Consequently, although $P(T \mid \text{do}(P))$ is not identifiable in general, the shard intervention $P(T \mid \text{do}(S = s))$ is identifiable under Model 1. Since S has no parents and no backdoor paths to T , do-calculus gives

$$P(T \mid \text{do}(S = s)) = P(T \mid S = s).$$

The audit therefore targets a quantity the auditor can actually estimate from samples, while the exclusion restriction ensures that any detected dependence between S and T is structurally attributable to the path $S \rightarrow P \rightarrow T$.

Notice that under Model 1, the causal model provenance problem becomes one-side testable. If there is no observed dependence between S and T , there are two possible cases. Either the candidate edge $P \rightarrow T$ is absent (so $T = f_T(C, U_T)$ does not depend on P at all), or the edge is present but f_T depends on P only through components that do not vary with S , e.g., inductive bias or outputs that are invariant to S . If there is an observed dependence between S and T , this corresponds to the case in which the edge is active *and* transmits shard signal along the path $S \rightarrow P \rightarrow T$. Consequently, significant dependence implies the edge exists, but failure to observe it does not imply its absence. A derivation that strips all shard-specific information from P before passing it to T is structurally indistinguishable, in this audit, from one that never used P .

Our framework therefore reduces model provenance to a membership-auditing-style question: can the auditor detect whether the target model T depends on the secret shard S ? A positive answer first establishes a statistical claim, namely that T is distinguishably dependent on S . The causal interpretation of this claim comes from Model 1: because the model rules out direct or confounded paths from S to T that bypass P , detected shard dependence is interpreted as evidence that T inherited shard-specific information through the candidate parent P .

Definition 2 formalizes the corresponding hypothesis testing problem. The auditor observes a tuple (S, S', T) , where S is the secret shard, S' is an independent control shard drawn from the same population, and T is the target model. The null hypothesis represents the case in which T is unrelated to S , while the alternative represents the case in which T is statistically dependent on S . To model the relationship between T and S , we define a conditional distribution for the target model given the shard. Under the null, the conditional distribution $\Pi_0 \mid S$ does not depend on S , so T is independent of the secret shard. Under the alternative, the conditional distribution $\Pi_1 \mid S$ depends on S , so T may contain shard-specific information.

Definition 2 (Hypothesis testing problem). For $b \in \{0, 1\}$, let $\Pi_b \mid S$ be a conditional distribution of the target model T given a shard S . The observation under Π_b is a tuple (S, S', T) , which is generated by

$$S, S' \sim \mathcal{D}^n, T \sim \Pi_b \mid S.$$

- Under the *null hypothesis* Π_0 , the conditional distribution $\Pi_0 \mid S$ does not depend on S , thus the target model T is independent of S .
- Under the *alternative hypothesis* Π_1 , the conditional distribution $\Pi_1 \mid S$ depends on S , thus the target model T is statistically dependent on S .

Definition 2 formulates the hypothesis testing problem at the level of joint distributions. In an audit, however, the auditor does not know the conditional distribution $\Pi_b \mid S$. Instead, the auditor has access to sampled shards S, S' and can query the target model T as a black box. Our key observation is that this dependence-testing problem can be reduced to a distinguishing game, thereby giving an operational test for the hypotheses in Definition 2.

We first describe the distinguishing game at a high level. The game is parameterized by an observation (S, S', T) and a distinguisher A . The distinguisher has access to the construction splits $S_{m+1:n}$

184 and $S'_{m+1:n}$ and black-box query access to the target model T . It is then evaluated on a single sam-
 185 ple x . In the *real* shard experiment, x is sampled from the held-out shard split $S_{:m}$; in the *ideal*
 186 shard experiment, x is sampled from the corresponding split $S'_{:m}$ of the independent control shard
 187 S' . The goal of the distinguisher is to decide whether its input came from the real or ideal shard
 188 experiment. We state the experiments and the corresponding distinguishing game in Definition 3
 189 and Definition 4.

190 **Definition 3** (Real and ideal shard experiments). Let $S, S' \sim \mathcal{D}^n$ independently and $T \sim \Pi \mid S$.
 191 Let A be a distinguisher that is given access to $S_{m+1:n}, S'_{m+1:n}$ and black-box query access to T .
 192 We define the real and ideal shard experiments with respect to (S, S', T) as

$$\text{Exp}_{T,S,S'}^{\text{real}}(A) : \begin{array}{l} x \xleftarrow{\$} S_{:m}; \\ \text{return } A(x). \end{array} \quad \text{Exp}_{T,S,S'}^{\text{ideal}}(A) : \begin{array}{l} x \xleftarrow{\$} S'_{:m}; \\ \text{return } A(x). \end{array}$$

193 **Definition 4** (Shard distinguishing game). Given an audit instance (S, S', T) and a distinguisher A
 194 constructed from $(T, S_{m+1:n}, S'_{m+1:n})$, the shard distinguishing game $\mathcal{D}_{T,S,S'}^{\text{shard}}(A)$ is defined as

195 $b \xleftarrow{\$} \{0, 1\}$; if $b = 1$ then $\hat{b} \leftarrow \text{Exp}_{T,S,S'}^{\text{real}}(A)$; if $b = 0$ then $\hat{b} \leftarrow \text{Exp}_{T,S,S'}^{\text{ideal}}(A)$; return 1 iff $\hat{b} = b$.

196 In the shard distinguishing game (Definition 4), the distinguishers form a class of binary-output
 197 algorithms. Each distinguisher $A \in \mathcal{A}$ is a (randomized) algorithm with black-box query access to
 198 the target model T and access to the construction splits $S_{m+1:n}$ and $S'_{m+1:n}$. Given a shard sample
 199 $x \in \mathcal{X}$, the distinguisher outputs a bit $A(x) \in \{0, 1\}$, where output 1 means that A predicts that x
 200 was generated by the real experiment, while output 0 means that A predicts that x was generated by
 201 the ideal experiment.

202 We quantify the power of a distinguisher A by its *shard advantage* in the shard distinguishing game.
 203 An advantage of 0 means that A cannot distinguish the real experiment from the ideal experiment
 204 and performs no better than random guessing. An advantage of 1 means that A perfectly identifies
 205 which experiment generated its input. The general case and formal definition is given in Definition 5.

206 **Definition 5** (Shard advantage). For a distinguisher A , define its shard advantage with respect to the
 207 observation (S, S', T) by

$$\text{Adv}_A(T, S, S') \stackrel{\text{def}}{=} 2 \left| \Pr [\mathcal{D}_{T,S,S'}^{\text{shard}}(A) = 1] - \frac{1}{2} \right|.$$

208 To see how the dependence-testing problem in Definition 2 reduces to the shard distinguishing game,
 209 observe that the shard advantage $\text{Adv}_A(T, S, S')$ serves as a test statistic. Under the null hypothesis,
 210 S and T are independent. Therefore, conditioning on the target model T and both construction splits
 211 $S_{m+1:n}$ and $S'_{m+1:n}$, the held-out shard split $S_{:m}$ and the control split $S'_{:m}$ are two independent
 212 samples from the same population distribution. Consequently, the signed difference underlying the
 213 shard advantage has expectation zero under the null, and any nonzero value of $\text{Adv}_A(T, S, S')$ arises
 214 from finite-sample fluctuation.

215 Proposition 1 formalizes this observation: under the null, every fixed distinguisher A constructed
 216 from $(T, S_{m+1:n}, S'_{m+1:n})$ has shard advantage at most $\sqrt{\frac{\log(2/\alpha)}{m}}$ with probability at least $1 -$
 217 α . Thus, a value of $\text{Adv}_A(T, S, S')$ that substantially exceeds this null fluctuation bound provides
 218 evidence that the real shard split is distinguishable from the independent control split, and therefore
 219 supports rejecting the null hypothesis.

220 **Proposition 1** (Shard advantage under the null; Proof is in Appendix C). *Under the null hypothesis*
 221 Π_0 of Definition 2, fix any distinguisher A constructed from $(T, S_{m+1:n}, S'_{m+1:n})$. For every $\alpha \in$
 222 $(0, 1)$, with probability at least $1 - \alpha$ over the draw of (S, S', T) ,

$$\text{Adv}_A(T, S, S') \leq \sqrt{\frac{\log(2/\alpha)}{m}}.$$

223 Obtaining a good test from the shard distinguishing game then reduces to finding a good distin-
 224 guisher A and estimating its shard advantage $\text{Adv}_A(T, S, S')$ accurately. The distinguisher is con-
 225 structed from the target model T and the construction splits $S_{m+1:n}$ and $S'_{m+1:n}$, while its advantage

Algorithm 1 Empirical advantage estimator $\text{Est}(A; T, S_{:m}, S'_{:m})$

- 1: **Input:** A distinguisher A , a target model T , the shard split $S_{:m}$, and the control split $S'_{:m}$
 - 2: **Output:** an estimate of $\text{Adv}_A(T, S, S')$
 - 3: Compute $a \leftarrow \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\}$
 - 4: Compute $a' \leftarrow \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\}$
 - 5: Return $|a - a'|$.
-

is estimated on the held-out shard split $S_{:m}$ and the independent control split $S'_{:m}$. Algorithm 1 defines an empirical estimator for the shard advantage of a fixed distinguisher.

Proposition 2 states that the empirical estimate from Algorithm 1 concentrates around the true shard advantage at the standard $O(1/\sqrt{m})$ rate.

Proposition 2 (Accuracy of the empirical advantage estimator; Proof is in Appendix C). *Let (T, S, S') be defined as above, and let A be a fixed, possibly randomized, distinguisher constructed from $(T, S_{m+1:n}, S'_{m+1:n})$. Suppose that separate calls to A use independent internal randomness. Then, for every $\alpha \in (0, 1)$, with probability at least $1 - \alpha$ over the internal randomness of the estimator Est ,*

$$|\text{Est}(A; T, S_{:m}, S'_{:m}) - \text{Adv}_A(T, S, S')| \leq \sqrt{\frac{\log(2/\alpha)}{m}}.$$

For the deterministic distinguisher used by the test, $\text{Est}(A; T, S_{:m}, S'_{:m}) = \text{Adv}_A(T, S, S')$ for every fixed audit instance, so there is no separate estimation-error term to add to the null fluctuation. Under Π_0 , conditional on $(T, S_{m+1:n}, S'_{m+1:n})$, the two held-out empirical averages are independent and have the same mean. Hoeffding’s inequality therefore gives

$$\Pr_{(S, S', T) \sim \Pi_0} [\text{Est}(A; T, S_{:m}, S'_{:m}) \geq \tau] \leq 2 \exp(-m\tau^2).$$

We thus reject when the empirical advantage exceeds $\gamma_\alpha = \sqrt{\log(2/\alpha)/m}$. Lemma 1 shows directly that this threshold controls the Type-I error at level α .

Lemma 1 (Level- α shard distinguishing test; Proof is in Appendix C). *Let $\alpha \in (0, 1)$ be a significance level and $\gamma_\alpha = \sqrt{\frac{\log(2/\alpha)}{m}}$. Let A be a deterministic distinguisher constructed from $(T, S_{m+1:n}, S'_{m+1:n})$, and define the thresholding test*

$$\phi_{A,\alpha}(T, S, S') \stackrel{\text{def}}{=} \mathbf{1}\{\text{Est}(A; T, S_{:m}, S'_{:m}) > \gamma_\alpha\}.$$

Type-I error. Under the null hypothesis Π_0 , $\Pr_{(S, S', T) \sim \Pi_0} [\phi_{A,\alpha}(T, S, S') = 1] \leq \alpha$.

Instance witness statement. When fixing any observation (T, S, S') , if $\text{Adv}_A(T, S, S') > \gamma_\alpha$, then,

$$\phi_{A,\alpha}(T, S, S') = 1.$$

Lemma 1 only establishes a statistical claim. Theorem 1 states the causal interpretation of the shard test under Model 1. Informally, the shard S has no direct path to T and no shard-specific confounding path that bypasses P , any detected dependence between S and T is interpreted as evidence that the candidate edge $P \rightarrow T$ is active. We note that the Type-I guarantee in Lemma 1 should be read as a guarantee against falsely detecting *shard dependence*, not as a standalone guarantee against every possible causal non-provenance world.

Theorem 1 (Provenance guarantee of the shard test; Proof is in Appendix C). *Under Model 1, let $\alpha \in (0, 1)$ and let $\phi_{A,\alpha}$ be the level- α shard distinguishing test from Lemma 1. In the no-provenance submodel in which the edge $P \rightarrow T$ is absent, (S, S', T) follows the null distribution Π_0 and*

$$\Pr_{(S, S', T) \sim \Pi_0} [\phi_{A,\alpha}(T, S, S') = 1] \leq \alpha.$$

For any fixed observation (T, S, S') , if

$$\text{Adv}_A(T, S, S') > \sqrt{\frac{\log(2/\alpha)}{m}},$$

then $\phi_{A,\alpha}(T, S, S') = 1$ and the test reports provenance. Thus, within Model 1, the probability of falsely reporting provenance is at most α ; failure to reject remains inconclusive.

Algorithm 2 Construction of the augmentation-based distinguisher

- 1: **Input:** A target model T , a fixed view family $\mathcal{G} = \{g_0, \dots, g_{K-1}\}$, a feature map Ψ , the construction split $S_{m+1:n}$, and the control construction split $S'_{m+1:n}$
- 2: **Output:** An augmentation-based distinguisher $A^{\text{aug}} : \mathcal{X} \rightarrow \{0, 1\}$
- 3: **for** each $x \in S_{m+1:n} \cup S'_{m+1:n}$ **do**
- 4: **for** $j \in \{0, \dots, K-1\}$ **do**
- 5: Compute the view $x_j \leftarrow g_j(x)$
- 6: Query the target model and compute $u_j \leftarrow T(x_j)$
- 7: Compute $z_T(x) \leftarrow \Psi(u_0, \dots, u_{K-1})$
- 8: Train a binary classifier $h_{\hat{\theta}}$ on

$$\{(z_T(x), 1) : x \in S_{m+1:n}\} \cup \{(z_T(x), 0) : x \in S'_{m+1:n}\}.$$

- 9: Define

$$A^{\text{aug}}(x) \stackrel{\text{def}}{=} h_{\hat{\theta}}(\Psi(T(g_0(x)), \dots, T(g_{K-1}(x)))).$$

- 10: Return A^{aug} .
-

259 **Construction of the distinguisher A .** The main advantage of our framework is that it only re-
260 quires a simple, non-trivial distinguisher. Thus, we can leverage existing distinguishers from the
261 membership inference literature [4, 10, 25, 26].

262 4 Experiments

263 In this section, we evaluate whether shard-based membership evidence only can be reliably used as
264 a practical signal for model provenance. We conduct the experiments for both ViT-like models and
265 LLMs. Our experiments are designed to answer the following questions

- 266 **(RQ1)** How effective is our testing framework at detecting shard signal in derived target models?
- 267 **(RQ2)** Does the advantage disappear under control settings where no membership signal exists?
- 268 **(RQ3)** How sensitive is our method to small secret-shard sizes?
- 269 **(RQ4)** Do all shards drawn from the parent model’s training data induce detectable membership
270 signals, and are such signals preserved in derived target models?

271 Throughout this section, m is the number of held-out examples in each of the two shards, so an
272 audit evaluates $2m$ examples in total. The level- α guarantee applies separately to each prespecified,
273 construction-trained distinguisher and assumes that the two held-out shards are independent samples
274 from the same population, as in Definition 2; we do not claim simultaneous family-wise error control
275 across table rows. To retain the direction of the empirical signal, the tables and figure report the
276 signed gap $\hat{\Delta} = a - a'$. The formal empirical advantage from Algorithm 1 is $\widehat{\text{Adv}} = |\hat{\Delta}|$, and every
277 test verdict compares $|\hat{\Delta}|$ with γ_α .

278 4.1 Vision Model Provenance

279 **Models.** We instantiate the audit with the parent model $P = \text{google/vit-base-patch16-224}$,
280 a ViT-B/16 image classifier pretrained on ImageNet-21k and fine-tuned on ImageNet-1k. Its pre-
281 training shard is therefore the ImageNet-1k training set, and the matched control distribution is the
282 ImageNet-1k validation set. We evaluate the test on seven candidate target models that declare P
283 as their base model on the HuggingFace Hub. We also include the trivial self-test $T = P$. We
284 further include two negative controls. The first replaces T ’s parameters with a fresh random ini-
285 tialisation of the same architecture as P , severing every path from the shard to T . The second is
286 facebook/deit-base-patch16-224, a ViT-B/16 trained *independently* on the same ImageNet-
287 1k corpus. The DeiT control is particularly informative because it satisfies the structural-equation
288 analogue of $S \leftarrow H \rightarrow T$ with H being the ImageNet itself. It probes the test’s robustness to the
289 shard-confounding clause of Model 1 in a regime where the model conditions are in fact violated.

Shard construction. We draw S and S' from the training and validation set of ImageNet-1k. We use the first 77 classes covered by every shard, and sample 25 images per class for the secret shard and 25 images per class for the control shard, giving $m = 1925$ records in each held-out shard ($2m = 3850$ evaluation records in total); the disjoint construction split has the same number per shard.

Distinguisher construction. For each held-out image x with ground-truth ImageNet class y , we define the per-image membership score

$$s_T(x, y) = \log \text{softmax}(P_{\text{head}}(T_{\text{enc}}(x)))_y,$$

where T_{enc} is the candidate target’s transformer encoder (its [CLS] token after the final layer) and P_{head} is the parent’s 1000-way classification head. Higher $s_T(x, y)$ means the parent’s head, applied to the target’s representation, assigns higher probability to the correct class. Here, routing through P_{head} rather than T ’s own (often task-mismatched) head gives a single shared score across all targets, including the random-initialisation control whose head is meaningless, and avoids conflating shard membership with task-specific calibration.

The distinguisher A is then the threshold rule $A(x) = \mathbf{1}\{s_T(x, y) > \tau\}$, with τ calibrated on the MIA train split by maximising balanced accuracy and evaluated on the disjoint MIA test split. Following Lemma 1 we report the signed gap $\hat{\Delta}_A(T, S, S')$, whose absolute value is the empirical estimate $\widehat{\text{Adv}}_A(T, S, S')$ from Algorithm 1, and reject the null whenever $|\hat{\Delta}_A(T, S, S')|$ exceeds $\gamma_{0.05} = \sqrt{\log(2/0.05)}/m = 0.044$ at $m = 1925$ per shard.

Table 1: Shard-membership auditing and provenance-test results on Vision Transformer target models. The level-0.05 test uses $\phi_{A,0.05} = \mathbf{1}\{|\hat{\Delta}_{\text{main}}| > \gamma_{0.05}\}$, where $\gamma_{0.05} = \sqrt{\log(2/0.05)}/m = 0.044$ for $m = 1925$ held-out examples in each shard. The candidate parent is the ImageNet-1k classifier google/vit-base-patch16-224; the secret shard S is sampled from ImageNet-1k train and the control shard S' from ImageNet-1k val, balanced across 77 classes (25 members + 25 non-members per class per split). The distinguisher uses the per-image score $\log P_{\text{head}}(T_{\text{enc}}(x))[y]$.

Parent model	Target model	Ctrl Acc.	$\hat{\Delta}_{\text{ctrl}}$	Main Acc.	$\hat{\Delta}_{\text{main}}$	$\gamma_{0.05}$	Verdict / Correct?
ViT-B/16-224 (google/vit-base-patch16-224)	google/vit-base-patch16-224 (self)	52.0%	0.041	52.4%	0.049	0.044	Reject H_0 / Yes
	ISx0din/vit-base-oxford-iiit-pets	52.0%	0.041	52.4%	0.049	0.044	Reject H_0 / Yes
	MatanBT/vit-base-patch16-224-cifar10	51.7%	0.034	53.4%	0.068	0.044	Reject H_0 / Yes
	MatanBT/vit-base-patch16-224-cifar100	50.0%	0.000	52.6%	0.053	0.044	Reject H_0 / Yes
	Abhiram4/vit-base-patch16-224-finetuned-eurosat	52.5%	0.051	52.3%	0.045	0.044	Reject H_0 / Yes
	MatanBT/vit-base-patch16-224-celeba-smiling	53.3%	0.066	52.3%	0.047	0.044	Reject H_0 / Yes
	rvv-karma/Human-Action-Recognition-ViT-Base-patch16-224	50.1%	0.002	53.0%	0.061	0.044	Reject H_0 / Yes
(negative controls)	ViT-B/16-224 random-init	50.4%	0.008	50.0%	-0.001	0.044	Fail to reject / Yes
	facebook/deit-base-patch16-224†	48.2%	-0.036	49.4%	-0.011	0.044	Fail to reject / Yes

4.2 Large Language Model Provenance

Models. We evaluate our method on both parent models and derived target models. Parent models are drawn from the open-source Pythia suite at different scales: 1B, 1.4B, 6.9B, and 12B [2]. Target models include fine-tuned Pythia checkpoints, including supervised fine-tuned models, instruction-tuned models, and preference-optimized models. This setting lets us test whether shard-specific information remains detectable after common post-training transformations.

Shard construction. For each audit instance, we construct a secret shard S by drawing randomly from the training dataset Pile [8] of the parent model, and the control set S' is drawn randomly from the validation set counterpart. For this purpose, we use the MIMIR dataset [6], which is constructed for membership inference attack settings. The dataset is decontaminated to remove duplicate samples from the training and testing sets. The resulting dataset has 1000 samples in the training and 740 samples in the validation set. We reserve 400 member examples and 400 non-member examples as the held-out evaluation split, so $m = 400$ per shard and the test uses $2m = 800$ held-out examples in total. The remaining 600 member examples and 340 non-member examples are used only for threshold calibration; their unequal sizes do not enter the held-out Hoeffding bound.

Distinguisher construction. We instantiate a simple distinguisher from the MIN-K [25] log-probability score with $k = 20\%$:

$$s_T^{(20)}(x) = \frac{1}{|\text{MinK}_{20}(x)|} \sum_{i \in \text{MinK}_{20}(x)} \log p_M(x_i | x_{<i}),$$

where $\text{MinK}_{20}(x)$ denotes the lowest-probability 20% of scored tokens. The distinguisher is then constructed by threshold calibration on the construction split only $A_{\tau_T^*, T}(x) = \mathbf{1}\{s_T^{(20)}(x) \geq \tau_T^*\}$, where τ_T^* is constructed from the construction splits.

Table 2: Shard-membership auditing and provenance-test results on fine-tuned Pythia target models with MIMIR GitHub benchmark. The level-0.05 test uses $\phi_{A,0.05} = \mathbf{1}\{|\hat{\Delta}_{\text{main}}| > \gamma_{0.05}\}$, where $\gamma_{0.05} = \sqrt{\log(2/0.05)/m} = 0.096$ for $m = 400$ held-out examples in each shard.

Parent model	Target model	Ctrl Acc.	$\hat{\Delta}_{\text{ctrl}}$	Main Acc.	$\hat{\Delta}_{\text{main}}$	$\gamma_{0.05}$	Verdict / Correct?
Pythia-1B	Leogrin Hh Sft	53.2%	0.065	69.5%	0.390	0.096	Reject H_0 / Yes
Pythia-1.4B	Hermaster1 4B Lamini Docs	52.2%	0.045	67.2%	0.345	0.096	Reject H_0 / Yes
	Kykim0 Tulu V2 Mix	52.0%	0.040	55.8%	0.115	0.096	Reject H_0 / Yes
	Linguacustodia Fin	52.0%	0.040	57.5%	0.150	0.096	Reject H_0 / Yes
	Lomahony Helpful Dpo	51.0%	0.020	66.2%	0.325	0.096	Reject H_0 / Yes
	Lomahony Helpful Sft	51.5%	0.030	66.0%	0.320	0.096	Reject H_0 / Yes
	Nnheui Sft Full	51.2%	0.025	68.8%	0.375	0.096	Reject H_0 / Yes
Pythia-6.9B	Allenai Tulu	51.5%	0.030	70.0%	0.400	0.096	Reject H_0 / Yes
	Lomahony Hh Dpo	52.5%	0.050	67.8%	0.355	0.096	Reject H_0 / Yes
	Lomahony Hh Sft	52.8%	0.055	69.0%	0.380	0.096	Reject H_0 / Yes
	Pkarypis Ultrachat	50.5%	0.010	69.8%	0.395	0.096	Reject H_0 / Yes
	Usvsnsp Ppo	51.7%	0.035	69.0%	0.380	0.096	Reject H_0 / Yes
Pythia-12B	Lomahony Hh Dpo	50.7%	0.015	70.0%	0.400	0.096	Reject H_0 / Yes
	Lomahony Hh Sft	48.0%	0.040	70.5%	0.410	0.096	Reject H_0 / Yes

Table 3: Parent-model shard-signal sanity check on the MIMIR GitHub benchmark. For each Pythia parent checkpoint, we test whether the MIN-K distinguisher detects pretraining membership signal between the MIMIR GitHub member shard S and matched nonmember shard S' . The level-0.05 reference threshold is $\gamma_{0.05} = \sqrt{\log(2/0.05)/m} = 0.096$ for $m = 400$ held-out examples in each shard.

Parent model	Ctrl Acc.	$\hat{\Delta}_{\text{ctrl}}$	Main Acc.	$\hat{\Delta}_{\text{main}}$	$\gamma_{0.05}$	Shard signal?
Pythia-1B	51.7%	0.035	67.5%	0.350	0.096	Yes
Pythia-1.4B	52.8%	0.055	69.0%	0.380	0.096	Yes
Pythia-6.9B	52.0%	0.040	69.0%	0.380	0.096	Yes
Pythia-12B	49.2%	0.015	70.5%	0.410	0.096	Yes

4.3 Results

(RQ1): Table 1 and 2 show that the shard-membership signal is consistently detectable in derived target models. Across all target checkpoints, the absolute main gap exceeds the threshold $\gamma_{0.05}$, so the test can reject the null hypothesis in every test case.

(RQ2): The control columns in Tables 2 and 4 remain below the threshold across different parents, targets, and shard domains, so the test correctly fails to reject shard dependence in these matched null comparisons.

(RQ3): Figure 1 shows how the provenance test behaves as the held-out shard size m shrinks for the CIFAR-10 fine-tune MatanBT/vit-base-patch16-224-cifar10. For each m , we draw 50 class-balanced subsamples from each of the construction and evaluation splits. We trace the random-initialisation negative control through the same pipeline so that the level- α behavior under the null can be checked at every m . The figure reports the seed-averaged signed gap and empirical rejection rate, together with the theoretical critical-threshold curve $\gamma_{0.05}(m)$.

(RQ4): Table 5 shows that not every shard is effective. Among the tested MIMIR domains, only GitHub produces a detectable parent-level signal under the MIN-K 20% distinguisher. Table 3 gives evidence that signal is preserved in the target.

Table 4: Shard-membership auditing results on swj0419/WikiMIA (WikiMIA_length32). The level-0.05 test uses $\phi_{A,0.05} = \mathbf{1}\{|\hat{\Delta}_{\text{main}}| > \gamma_{0.05}\}$, where $\gamma_{0.05} = \sqrt{\log(2/0.05)/m} = 0.136$ for $m = 200$ held-out examples in each shard. Control columns report the nonmember-vs-nonmember null advantage.

Model	Score	Ctrl Acc.	$\hat{\Delta}_{\text{ctrl}}$	Main Acc.	$\hat{\Delta}_{\text{main}}$	$\gamma_{0.05}$	Verdict / Correct?
OPT-6.7B	Mean log-prob.	52.5%	0.070	60.8%	0.215	0.136	Reject H_0 / Yes
	Min-K% ($k = 5$)	49.5%	-0.010	57.5%	0.150	0.136	Reject H_0 / Yes
	Min-K% ($k = 10$)	49.0%	-0.020	61.0%	0.220	0.136	Reject H_0 / Yes
	Min-K% ($k = 20$)	52.5%	0.050	63.0%	0.260	0.136	Reject H_0 / Yes
	Min-K% ($k = 40$)	52.0%	0.040	59.0%	0.180	0.136	Reject H_0 / Yes
Pythia-6.9B	Mean log-prob.	55.0%	0.100	65.2%	0.305	0.136	Reject H_0 / Yes
	Min-K% ($k = 5$)	51.0%	0.020	59.5%	0.190	0.136	Reject H_0 / Yes
	Min-K% ($k = 10$)	52.5%	0.050	64.0%	0.280	0.136	Reject H_0 / Yes
	Min-K% ($k = 20$)	48.0%	-0.040	65.0%	0.300	0.136	Reject H_0 / Yes
	Min-K% ($k = 40$)	47.5%	-0.050	65.5%	0.310	0.136	Reject H_0 / Yes

Table 5: Multi-domain shard-signal sanity check on MIMIR for $P = \text{EleutherAI/pythia-1.4b}$ and $T = \text{nnheui/pythia-1.4b-sft-full}$. For each domain, we test whether the MIN-K 20% distinguisher detects pretraining membership signal in the parent and whether the signal remains in the target. The distinguisher threshold is calibrated on 300 construction examples per class and evaluated on 400 held-out examples per shard. The level-0.05 reference threshold is $\gamma_{0.05} = \sqrt{\log(2/0.05)/m} = 0.096$ for $m = 400$ held-out examples per shard. Except for GitHub, the parent signal is below the threshold, so these domains are not reliable provenance shards under this scalar distinguisher.

Domain	$\hat{\Delta}_P$	Parent verdict / Correct?	$\hat{\Delta}_T$	Target verdict / Correct?	Shard signal?
arxiv	-0.015	Fail to reject H_0 / No	0.005	Fail to reject H_0 / No	No
dm_mathematics	0.070	Fail to reject H_0 / No	0.062	Fail to reject H_0 / No	No
github	0.387	Reject H_0 / Yes	0.365	Reject H_0 / Yes	Yes
pile_cc	-0.025	Fail to reject H_0 / No	-0.002	Fail to reject H_0 / No	No
wikipedia_en	0.003	Fail to reject H_0 / No	0.017	Fail to reject H_0 / No	No

5 Conclusion

In this work, we propose a causal model for black-box model provenance that aligns the test statistic with the reason provenance matters: the inheritance of specific secret shard information from a candidate parent. By treating an auditor-chosen secret shard as an instrumental variable, we reduce provenance testing to a shard-membership distinguishing game. We obtain a finite-sample hypothesis test with a fixed false positive rate. Under the structural conditions of our model, rejection of the null supports a provenance claim from parent to target with a bounded false-positive rate. Instantiating the framework for diverse sets of models shows the effectiveness of our estimator.

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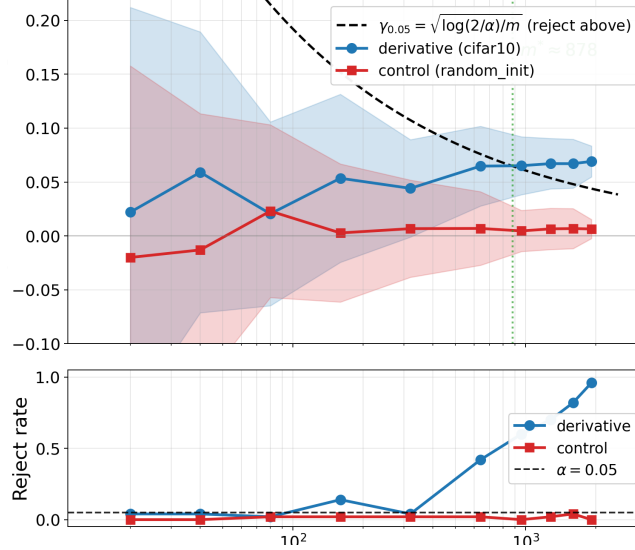


Figure 1: Audit performance versus held-out size m per shard; the dashed curve is the positive boundary of the two-sided test.

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488 A Detailed Related Work

489 **Model provenance.** Zhu et al. [37] approach a closely related but distinct question of whether two
 490 open-weight models were trained independently, by exploiting weight and activation symmetries of
 491 the LLM architectures. Independence testing certifies the absence of a derivation relation, while
 492 our test is structured to detect its presence with a controlled false-positive rate. We also distinguish
 493 provenance from *training-data* provenance verification for synthetic data [29], where the question
 494 is whether a downstream model was trained on outputs of a particular generative model rather than
 495 whether a downstream model is a derivative of a particular parent model.

496 **Model extraction and intellectual-property protection.** Adjacent to provenance is the literature
 497 on model extraction. Tramèr et al. [28] show that prediction APIs can be queried to extract model pa-
 498 rameters or functionality, Orekondy et al. [18] train “knockoff” substitutes from black-box queries,
 499 and Carlini et al. [5] extract pieces of production LMs through API queries. These attacks motivate
 500 provenance as a defense and accountability mechanism: once a derivative model is suspected of
 501 being trained from extracted outputs, the IP holder needs a statistical means of substantiating that
 502 claim against an uncooperative respondent.

503 **Causal identification with instrumental variables.** Methodologically, our analysis builds on
 504 causal inference and graphical models [20, 27]. The lineage relation between parent and target
 505 is confounded by background factors that are typically unobserved (architecture choice, pretrain-
 506 ing corpora, fine-tuning recipes, regularization), making the derivation effect non-identifiable from
 507 black-box outputs alone. We resolve this by using the secret shard as an instrumental variable in the
 508 sense of Angrist et al. [1]. Under our structural model, the shard influences the target only via the
 509 parent, so a non-trivial shard signal in the target identifies a derivation effect. While instrumental-
 510 variable arguments are standard in econometrics and epidemiology, their use to formalize black-box
 511 provenance is, to our knowledge, new.

512 B Limitations

513 The causal interpretation of our framework depends on the structural causal model in Model 1. In
 514 particular, the secret shard S must influence the target model T only through the candidate parent
 515 P , and there must be no shard-specific confounding path that bypasses P . The model conditions
 516 can be violated if the shard appears in another public or private corpus used by the target, or if data
 517 leakage creates a direct path from S to T . In such cases, the dependence testing evidence would no
 518 longer be able to justify the provenance interpretation $P \rightarrow T$.

519 Our empirical evaluation is limited in scope, and the power of our test may vary across model
 520 families, data domains, languages, post-training methods, and shard sizes. In particular, the power
 521 of the test depends on the choice of shard S , the quality of the matched control shard S' , the amount
 522 of held-out data m , and the effectiveness of the chosen distinguisher A .

523 C Proofs for Section 3

524 *Proof of Proposition 1.* Under the null hypothesis Π_0 , the conditional distribution $\Pi_0 \mid S$ does not
 525 depend on S , thereby the target model T is independent of the shard S . Let $(x_1, \dots, x_m) = S_{:m}$
 526 By the definition of the real and ideal shard experiments, we then have

$$\begin{aligned}
 \Pr \left[\text{Exp}_{T,S,S'}^{\text{real}}(A) = 1 \right] &= \Pr [A(x_I) = 1] && (I \stackrel{\$}{\leftarrow} \{1, \dots, m\}) \\
 &= \sum_{i=1}^m \Pr [I = i] \Pr [A(x_i) = 1 \mid I = i] \\
 &= \sum_{i=1}^m \frac{1}{m} \Pr [A(x_i) = 1] \\
 &= \frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1],
 \end{aligned}$$

527 and

$$\Pr \left[\text{Exp}_{T,S,S'}^{\text{ideal}}(A) = 1 \right] = \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1].$$

528 Plugging $\Pr \left[\text{Exp}_{T,S,S'}^{\text{real}}(A) = 1 \right]$ and $\Pr \left[\text{Exp}_{T,S,S'}^{\text{ideal}}(A) = 1 \right]$ into the expression of
 529 $\text{Adv}_A(T, S, S')$, we have

$$\text{Adv}_A(T, S, S') = \left| \frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1] - \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1] \right|.$$

530 Conditional on $(T, S_{m+1:n}, S'_{m+1:n})$, the samples $x_1, \dots, x_m$ and $x'_1, \dots, x'_m$ are indepen-
 531 dent draws from $\mathcal{D}$, thereby the random variables $\Pr [A(x_1) = 1], \dots, \Pr [A(x_m) = 1]$ and
 532 $\Pr [A(x'_1) = 1] \dots \Pr [A(x'_m) = 1]$ are independent and bounded in $[0, 1]$.

533 Moreover, for every $i, j \in [m]$,

$$\mathbf{Ex} [\Pr [A(x_i) = 1] \mid T, S_{m+1:n}, S'_{m+1:n}] = \mathbf{Ex} [\Pr [A(x'_j) = 1] \mid T, S_{m+1:n}, S'_{m+1:n}],$$

534 thus we have

$$\mathbf{Ex} \left[\frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1] - \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1] \mid T, S_{m+1:n}, S'_{m+1:n} \right] = 0.$$

535 Applying Hoeffding's inequality to the independent bounded summands $\frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1]$
 536 and $-\frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1]$ gives, for every $\tau > 0$,

$$\Pr [\text{Adv}_A(T, S, S') \geq \tau \mid T, S_{m+1:n}, S'_{m+1:n}] \leq 2 \exp(-m\tau^2).$$

537 Since this bound holds for every realization of $(T, S_{m+1:n}, S'_{m+1:n})$, it also holds unconditionally:

$$\Pr [\text{Adv}_A(T, S, S') \geq \tau] \leq 2 \exp(-m\tau^2).$$

538 Setting

$$\tau = \sqrt{\frac{\log(2/\alpha)}{m}}$$

539 yields

$$\Pr \left[\text{Adv}_A(T, S, S') \leq \sqrt{\frac{\log(2/\alpha)}{m}} \right] \geq 1 - \alpha,$$

540 which gives the desired statement. $\square$

541 *Proof of Proposition 2.* Let (T, S, S') be the observation and denote by

$$S_{:m} = (x_1, \dots, x_m), \quad S'_{:m} = (x'_1, \dots, x'_m).$$

542 Let A be the fixed distinguisher constructed from $(T, S_{m+1:n}, S'_{m+1:n})$. By the definition of the real
 543 and ideal shard experiments, we have

$$\Pr \left[\text{Exp}_{T,S,S'}^{\text{real}}(A) = 1 \right] = \frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1],$$

544 where the probabilities are over the internal randomness of A . Similarly,

$$\Pr \left[\text{Exp}_{T,S,S'}^{\text{ideal}}(A) = 1 \right] = \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1].$$

545 Plugging these two expressions into the definition of $\text{Adv}_A(T, S, S')$, we get

$$\text{Adv}_A(T, S, S') = \left| \frac{1}{m} \sum_{i=1}^m \Pr[A(x_i) = 1] - \frac{1}{m} \sum_{i=1}^m \Pr[A(x'_i) = 1] \right|.$$

546 On the other hand, the estimator $\text{Est}(A; T, S_{:m}, S'_{:m})$ computes

$$\text{Est}(A; T, S_{:m}, S'_{:m}) = \left| \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\} - \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\} \right|.$$

547 Define the signed quantities

$$\begin{aligned} \Delta_A(T, S, S') &\stackrel{\text{def}}{=} \frac{1}{m} \sum_{i=1}^m \Pr[A(x_i) = 1] - \frac{1}{m} \sum_{i=1}^m \Pr[A(x'_i) = 1], \\ \widehat{\Delta}_A(T, S, S') &\stackrel{\text{def}}{=} \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\} - \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\}. \end{aligned}$$

548 Then

$$\text{Adv}_A(T, S, S') = |\Delta_A(T, S, S')|, \quad \text{Est}(A; T, S_{:m}, S'_{:m}) = \left| \widehat{\Delta}_A(T, S, S') \right|.$$

549 For every $i \in [m]$, we have

$$\mathbf{E} \mathbf{x} [\mathbf{1}\{A(x_i) = 1\}] = \Pr[A(x_i) = 1], \quad \mathbf{E} \mathbf{x} [\mathbf{1}\{A(x'_i) = 1\}] = \Pr[A(x'_i) = 1].$$

550 Therefore,

$$\mathbf{E} \mathbf{x} [\widehat{\Delta}_A(T, S, S')] = \Delta_A(T, S, S').$$

551 Moreover, when each invocation of A uses fresh independent randomness, the random variables
552 $\mathbf{1}\{A(x_1) = 1\}, \dots, \mathbf{1}\{A(x_m) = 1\}$, and $\mathbf{1}\{A(x'_1) = 1\}, \dots, \mathbf{1}\{A(x'_m) = 1\}$ are independent and
553 bounded in $[0, 1]$. Applying Hoeffding's inequality to the independent bounded summands in

$$\widehat{\Delta}_A(T, S, S') = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\} - \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\}$$

554 gives, for every $\tau > 0$,

$$\Pr \left[\left| \widehat{\Delta}_A(T, S, S') - \Delta_A(T, S, S') \right| \geq \tau \right] \leq 2 \exp(-m\tau^2).$$

555 Since the absolute-value map is 1-Lipschitz, we have

$$\begin{aligned} |\text{Est}(A; T, S_{:m}, S'_{:m}) - \text{Adv}_A(T, S, S')| &= \left| \left| \widehat{\Delta}_A(T, S, S') \right| - |\Delta_A(T, S, S')| \right| \\ &\leq \left| \widehat{\Delta}_A(T, S, S') - \Delta_A(T, S, S') \right|. \end{aligned}$$

556 Therefore,

$$\Pr[|\text{Est}(A; T, S_{:m}, S'_{:m}) - \text{Adv}_A(T, S, S')| \geq \tau] \leq 2 \exp(-m\tau^2).$$

557 Setting

$$\tau = \sqrt{\frac{\log(2/\alpha)}{m}}$$

558 yields the desired statement. $\square$

559 *Proof of Lemma 1.* We first prove the Type-I error bound. Under the null hypothesis Π_0 , the target
 560 model T is independent of the secret shard S . Conditioned on $(T, S_{m+1:n}, S'_{m+1:n})$, the distin-
 561 guisher A is fixed, and the held-out samples

$$S_{:m} = (x_1, \dots, x_m), \quad S'_{:m} = (x'_1, \dots, x'_m)$$

562 are independent samples from the same population distribution. Since A is deterministic, the random
 563 variables

$$\mathbf{1}\{A(x_i) = 1\}, \quad \mathbf{1}\{A(x'_i) = 1\}$$

564 are independent, bounded in $[0, 1]$, and have the same conditional mean under the null.

565 By the definition of Est , we can write

$$\text{Est}(A; T, S_{:m}, S'_{:m}) = \left| \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\} - \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\} \right|.$$

566 Applying Hoeffding's inequality to the signed difference of the two empirical averages gives, for
 567 every $\tau > 0$,

$$\Pr [\text{Est}(A; T, S_{:m}, S'_{:m}) \geq \tau \mid T, S_{m+1:n}, S'_{m+1:n}] \leq 2 \exp(-m\tau^2).$$

568 Since this bound holds for every realization of $(T, S_{m+1:n}, S'_{m+1:n})$, it also holds unconditionally.
 569 This is a direct concentration bound on the statistic used by the test, so no second deviation term or
 570 union bound is needed. Setting

$$\tau = \gamma_\alpha = \sqrt{\frac{\log(2/\alpha)}{m}}$$

571 yields

$$\Pr_{(S, S', T) \sim \Pi_0} [\text{Est}(A; T, S_{:m}, S'_{:m}) > \gamma_\alpha] \leq \alpha.$$

572 By the definition of the thresholding test, this is exactly

$$\Pr_{(S, S', T) \sim \Pi_0} [\phi_{A, \alpha}(T, S, S') = 1] \leq \alpha.$$

573 We now prove the instance witness statement. Fix an arbitrary observation (T, S, S') . Since A is
 574 deterministic, the probabilities in the real and ideal shard experiments are only over the uniform
 575 choice of a held-out sample. Therefore,

$$\Pr [\text{Exp}_{T, S, S'}^{\text{real}}(A) = 1] = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\},$$

576 and similarly,

$$\Pr [\text{Exp}_{T, S, S'}^{\text{ideal}}(A) = 1] = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\}.$$

577 Hence

$$\text{Adv}_A(T, S, S') = \text{Est}(A; T, S_{:m}, S'_{:m}).$$

578 Thus, if $\text{Adv}_A(T, S, S') > \gamma_\alpha$, then

$$\text{Est}(A; T, S_{:m}, S'_{:m}) > \gamma_\alpha.$$

579 By the definition of $\phi_{A, \alpha}$, we conclude that

$$\phi_{A, \alpha}(T, S, S') = 1.$$

580 □

581 *Proof of Theorem 1.* First consider the no-provenance submodel in which the edge $P \rightarrow T$ is ab-
 582 sent. Under Model 1, S is exogenous, there is no direct edge $S \rightarrow T$, and there is no shard-specific
 583 confounding path from S to T . With the mediated path through P also absent, S and T are inde-
 584 pendent, so (S, S', T) follows the null distribution Π_0 . The Type-I statement of Lemma 1 therefore
 585 gives

$$\Pr_{(S, S', T) \sim \Pi_0} [\phi_{A, \alpha}(T, S, S') = 1] \leq \alpha.$$

586 Thus the probability of falsely reporting provenance is at most α within the stated causal model.

587 Now fix an observation (T, S, S') satisfying

$$\text{Adv}_A(T, S, S') > \sqrt{\frac{\log(2/\alpha)}{m}}.$$

588 By the instance witness statement in Lemma 1, $\phi_{A, \alpha}(T, S, S') = 1$, so the test reports provenance.
 589 Under the exclusion and no-shard-specific-confounding conditions of Model 1, genuine dependence
 590 between S and T can only be transmitted along $S \rightarrow P \rightarrow T$. The report is therefore level- α
 591 evidence for the provenance edge; the lemma does not assert a separate power guarantee. $\square$

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